



The Future of

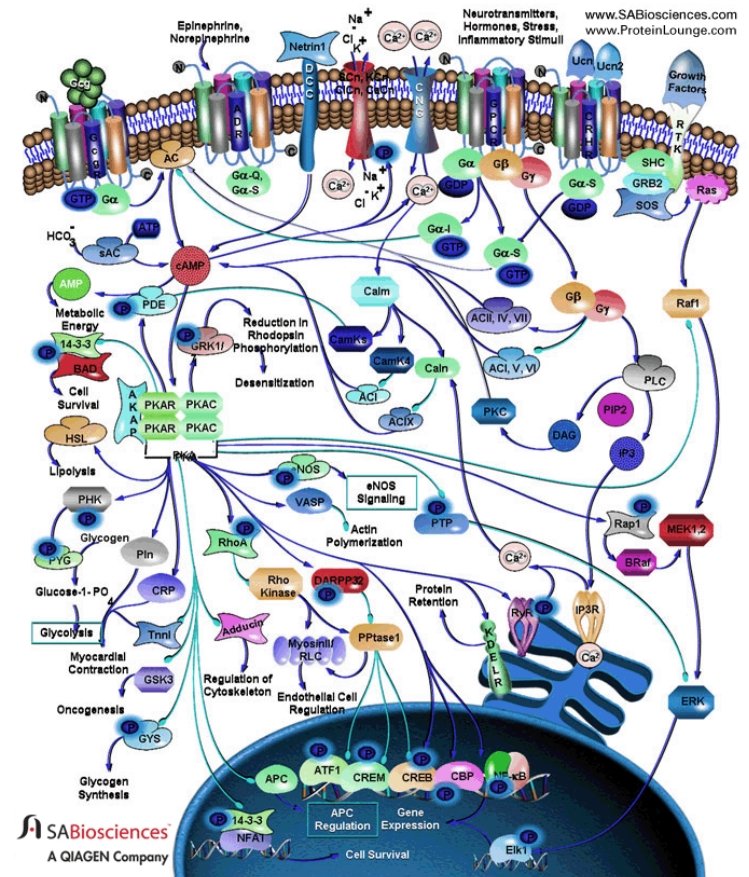
**Cancer diagnosis, prognosis
and therapy**

Aleksandra Filipovic
MD PhD

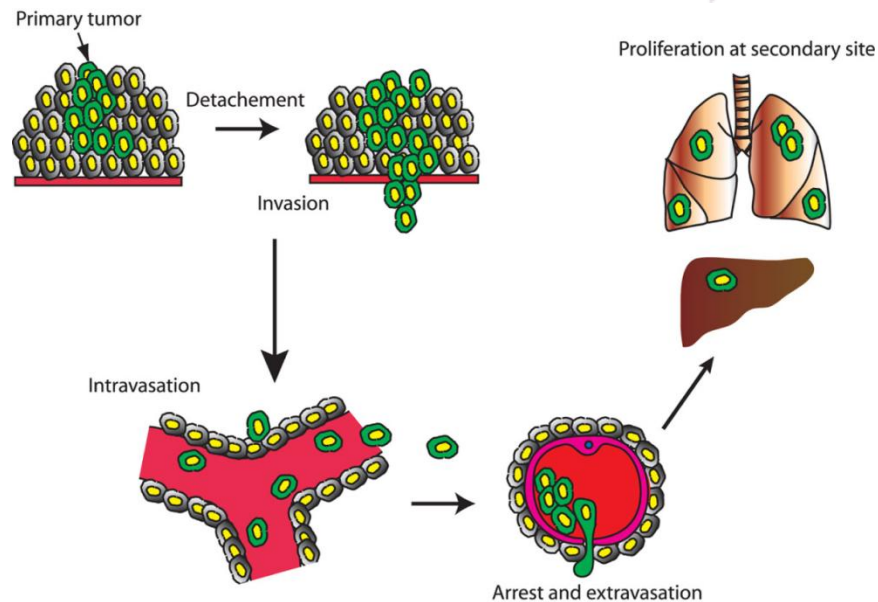
Overview

- **Complexity of cancer**
- **Challenges of translational research**
- **Breakthrough of genetic testing**
- **Targeted therapy and personalized medicine**
- **Cost effectiveness**
- **Conclusions**

Not a single target

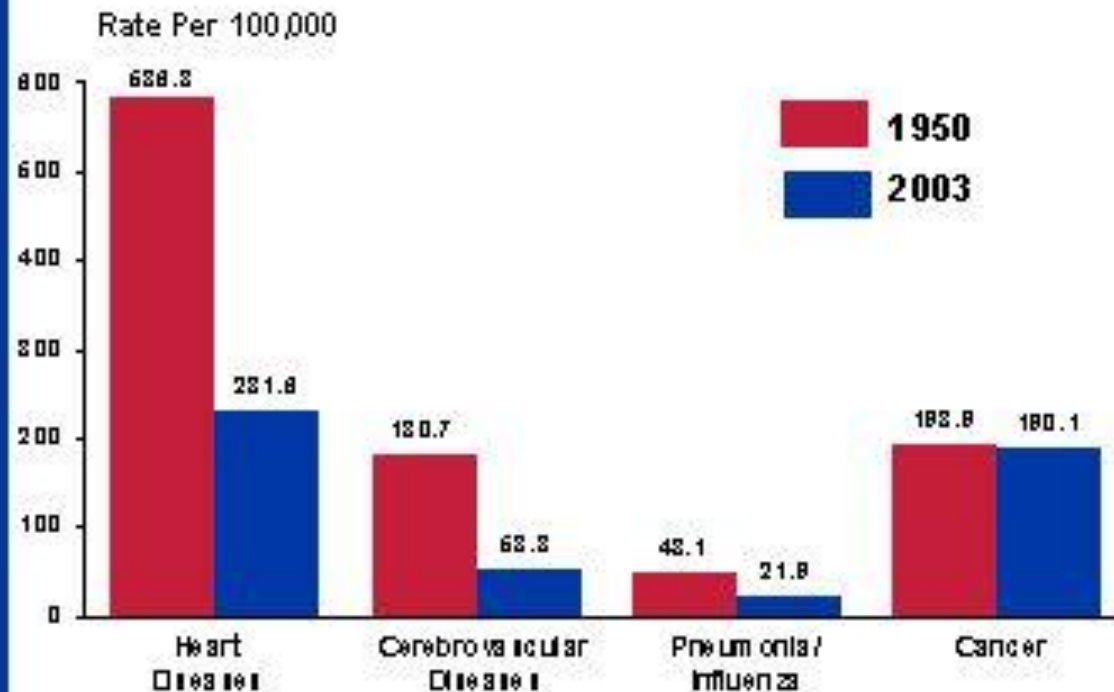


Complexity of cancer



Cancer Death Rates

Change in the US Death Rates* by Cause,
1950 & 2003

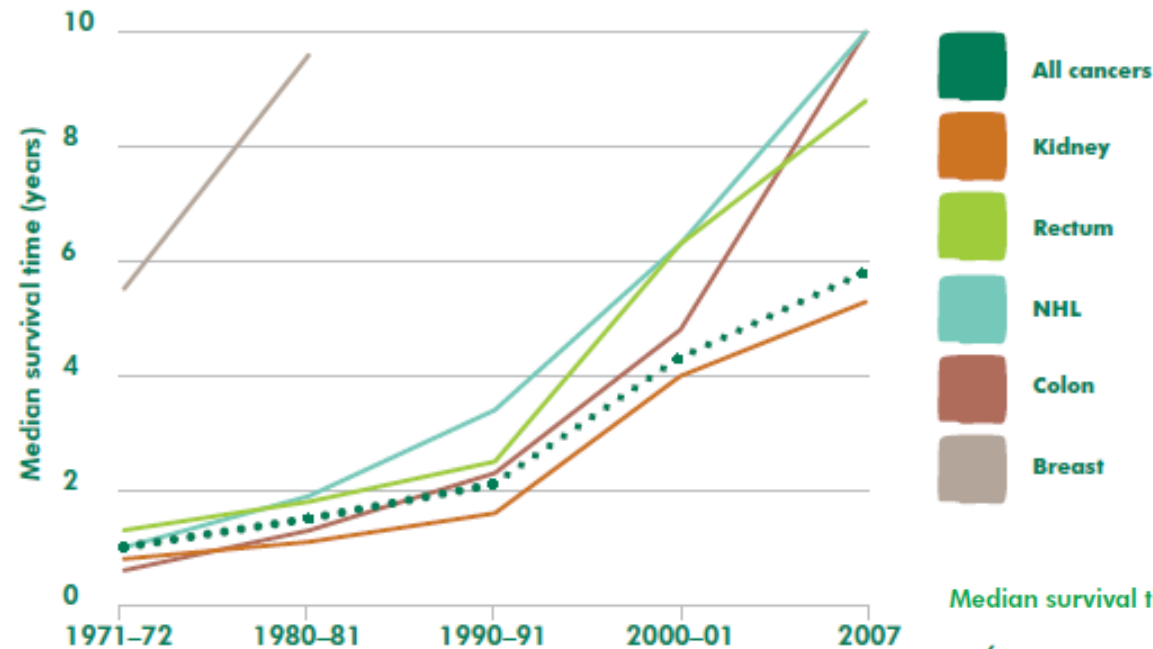


* Age-adjusted to 2000 US standard population.

Sources: 1950 Mortality Data - CDC/NCHS, NCHS Mortality Research

2003 Mortality Data - US Mortality Public Use Data (Apr. 2003), NCHS, Center for Disease Control and Prevention, 2003

Median survival time (years), by period of diagnosis and cancer¹

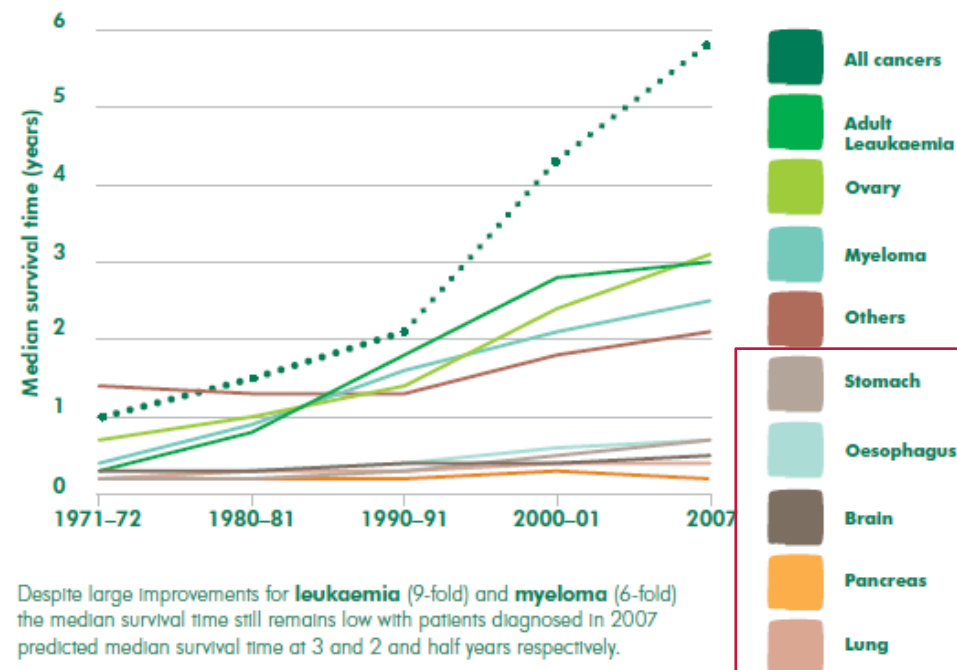


40 years ago median survival was 1yr while today it is 6 yrs

-Progress in breast, NHL, Colon

-Still poor survival for pancreas, squamous cell, lung CA

Median survival time (years), by period of diagnosis and cancer¹



Despite large improvements for **leukaemia** (9-fold) and **myeloma** (6-fold) the median survival time still remains low with patients diagnosed in 2007 predicted median survival time at 3 and 2 and half years respectively.

Can we accurately predict survival?

The recent National Institutes of Health Consensus Conference on improving end-of-life care concluded that it is currently not possible to accurately predict the timing of an individual's death.¹

- **Clinical prediction of survival influences treatment**
- **Options for treating advanced cancer are expanding**
- **Patients base their choices on prognosis**

Can we accurately predict survival?

The recent National Institutes of Health Consensus Conference on improving end-of-life care concluded that it is currently not possible to accurately predict the timing of an individual's death.¹

- **Twice as likely to be overly optimistic**
- **Correct within a week in 25%**
- **Off by more than a month in > 25%**
- **Despite inaccuracy, there was strong correlation between predicted and actual survival up to 6 months**
- **Predictions of less than 4 weeks were most precise ('horizon effect')**
- **No relation to actual survival for predictions beyond 6 months**

Circulating tumour cells



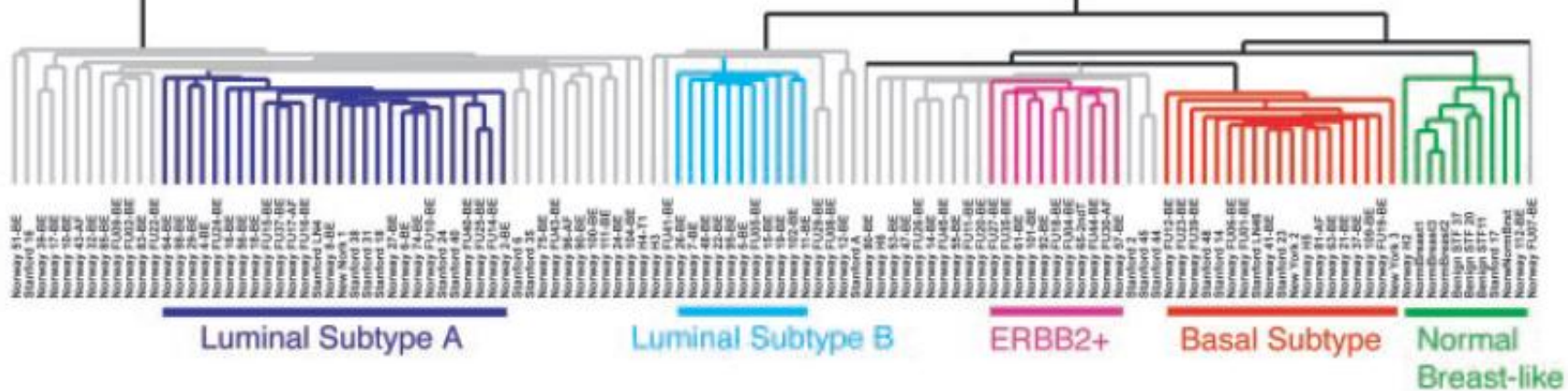
Possible Prognostic Factors in Patients With Advanced Cancer

TUMOR-RELATED	PATIENT-RELATED	ENVIRONMENTAL
Anatomic extent	Age, gender, race	Marital status, social support
Histologic grade and other features	Genetic make up	Geographic location
Tumor bulk	Performance status	Access to treatment
Tumor marker level	Symptoms and signs	Expertise of oncologist
Tumor molecular biology	Psychologic status	Response to treatment
Disease-free interval	Function of vital organs	Oncologist's prediction of survival
	Comorbidities	

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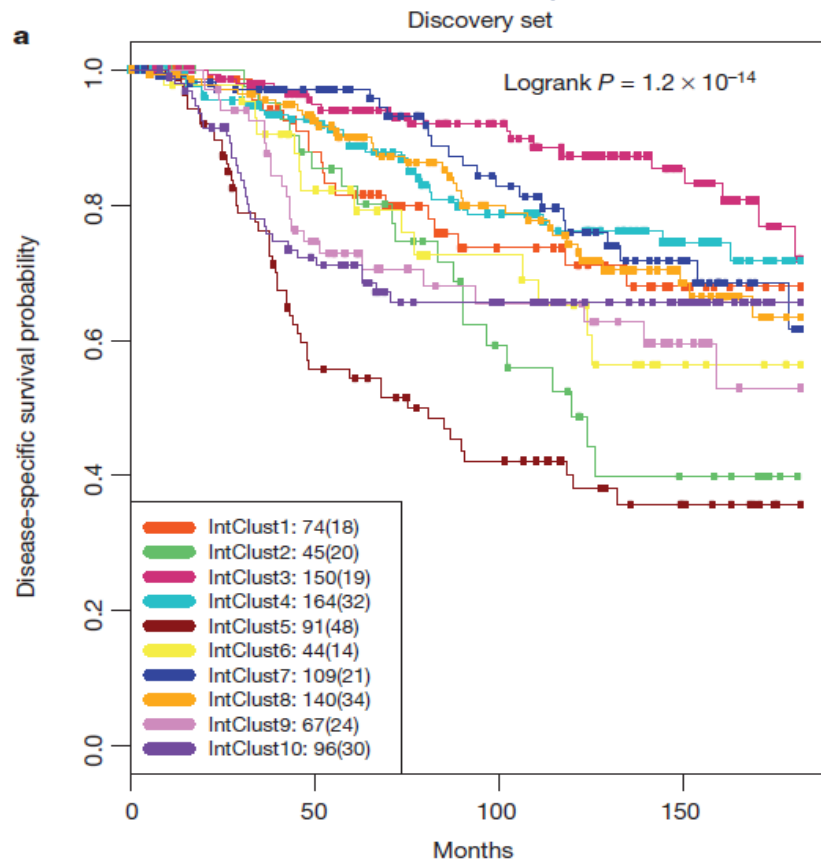
THERE IS MORE TO IT THAN MEETS THE EYE

Five main breast cancer subtypes

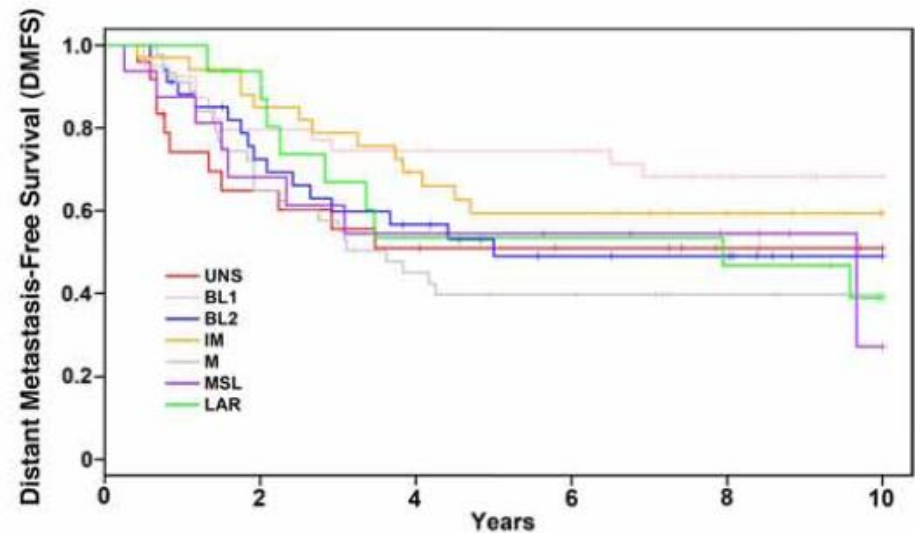


10 molecular subtypes

a



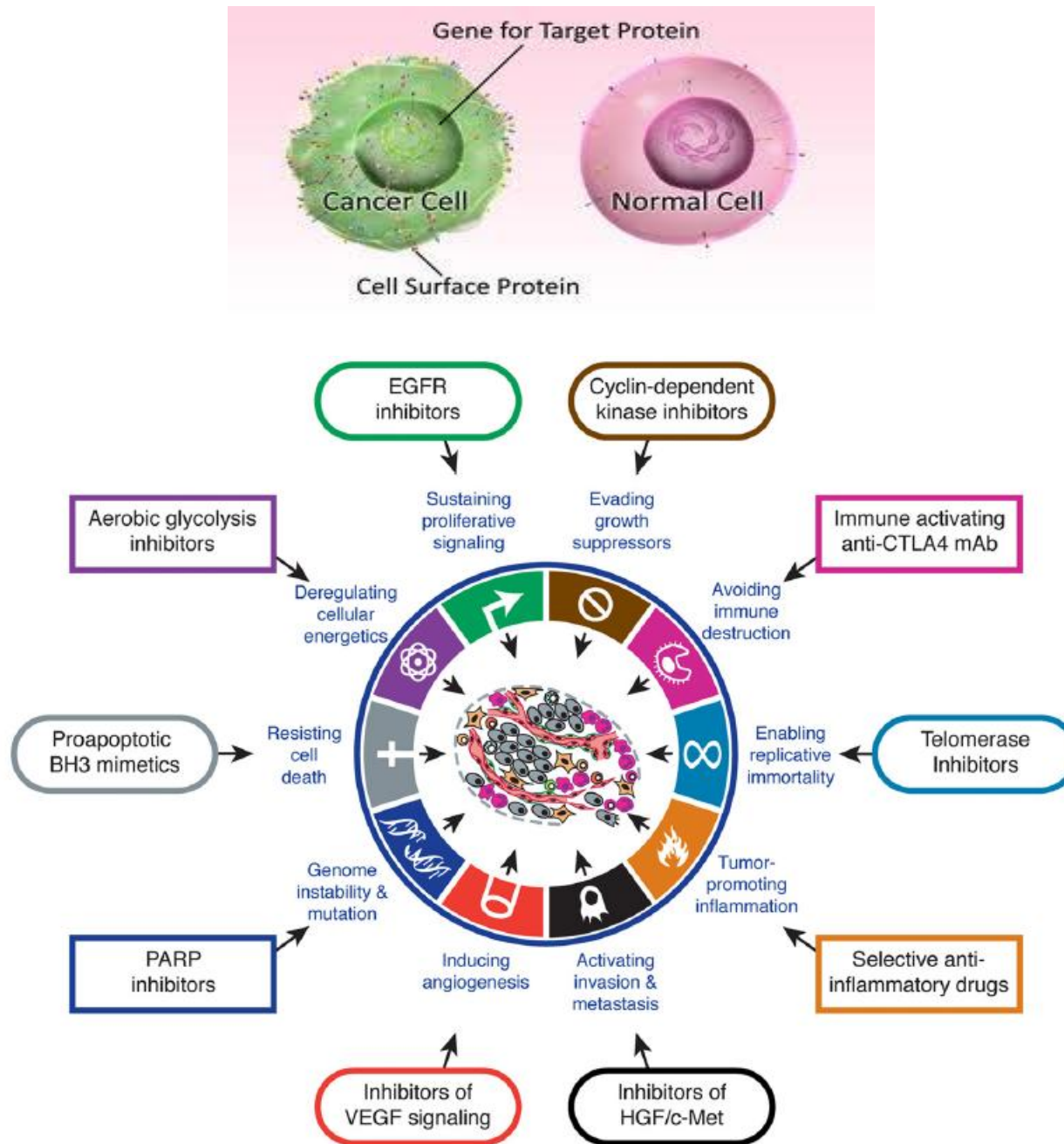
Seven Triple Negative Subtypes



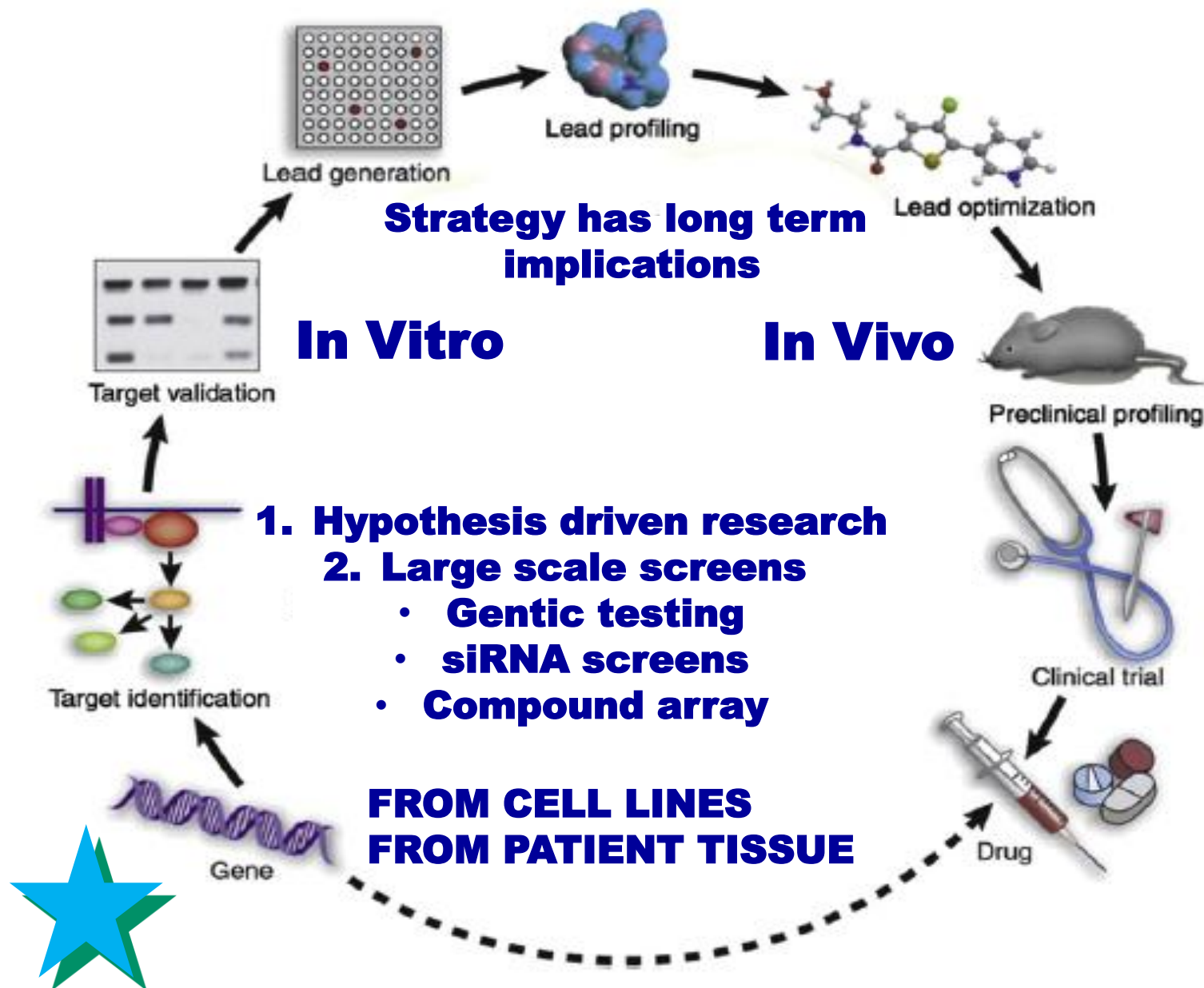
Cancer treatments: are we aiming at the right target?



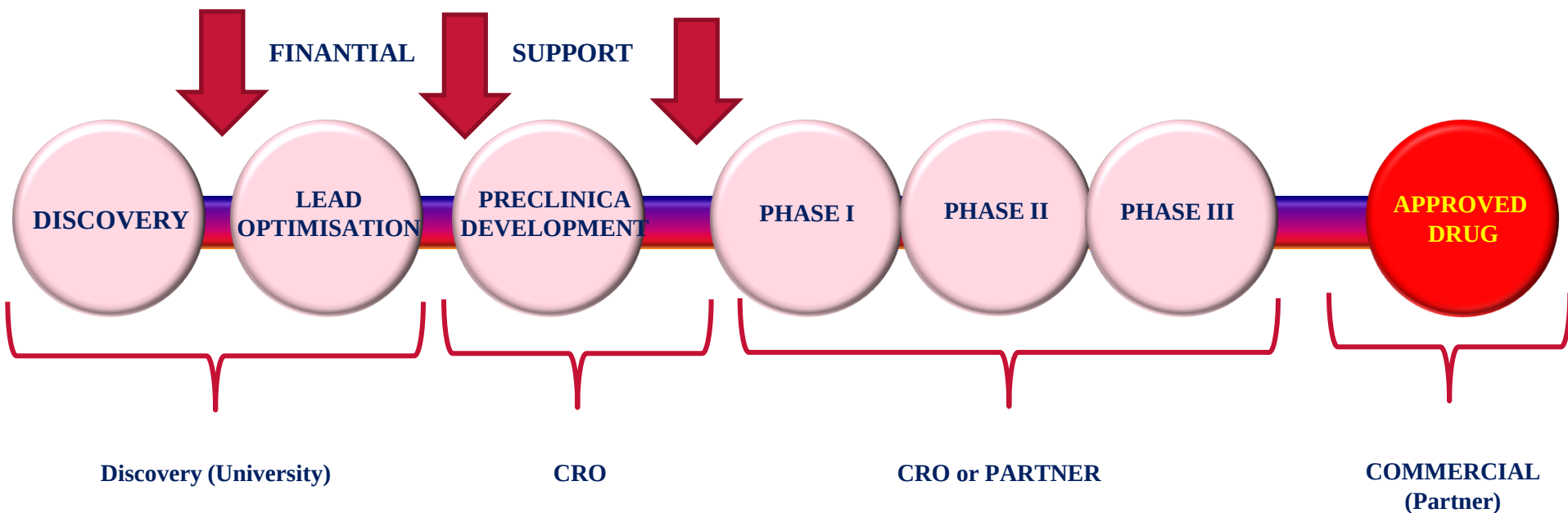
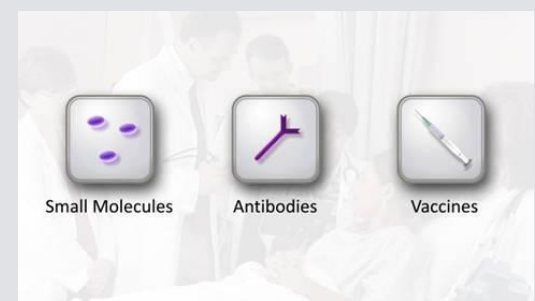
What is a cancer therapeutic target ?



The Long Road Home: Target/Drug Discovery and Development



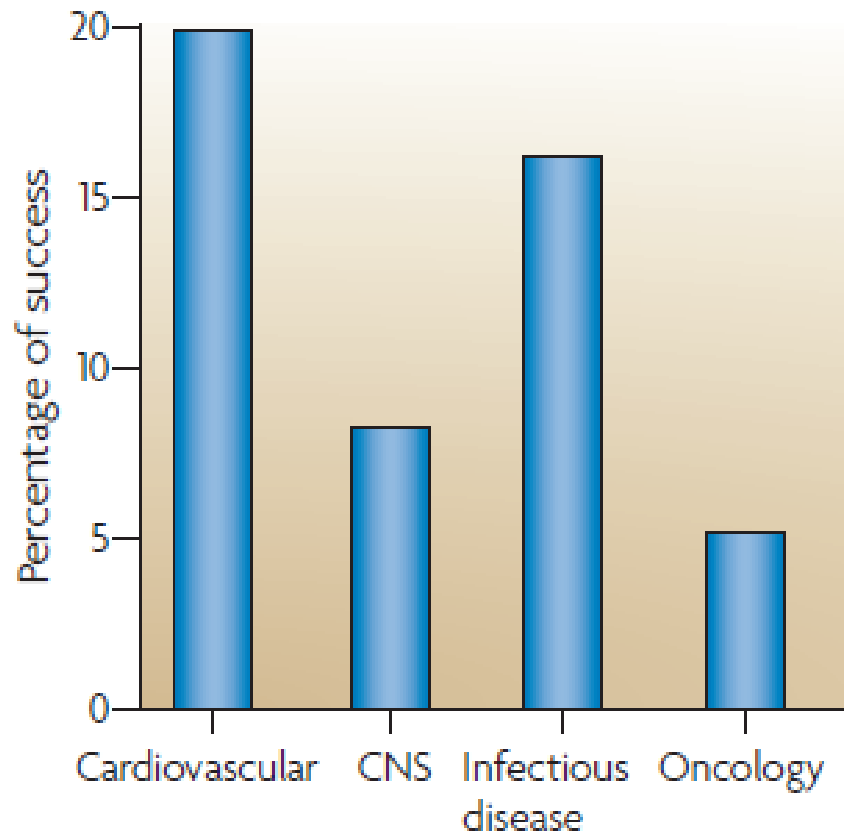
Types of therapeutics



- 1. Time consuming**
- 2. Costly**
- 3. Risk of failure**

PERCENT FAIL RATE IN PHASE I TRIALS IS 85%

DRUG APPROVAL RATE



PATIENT RECRUITMENT

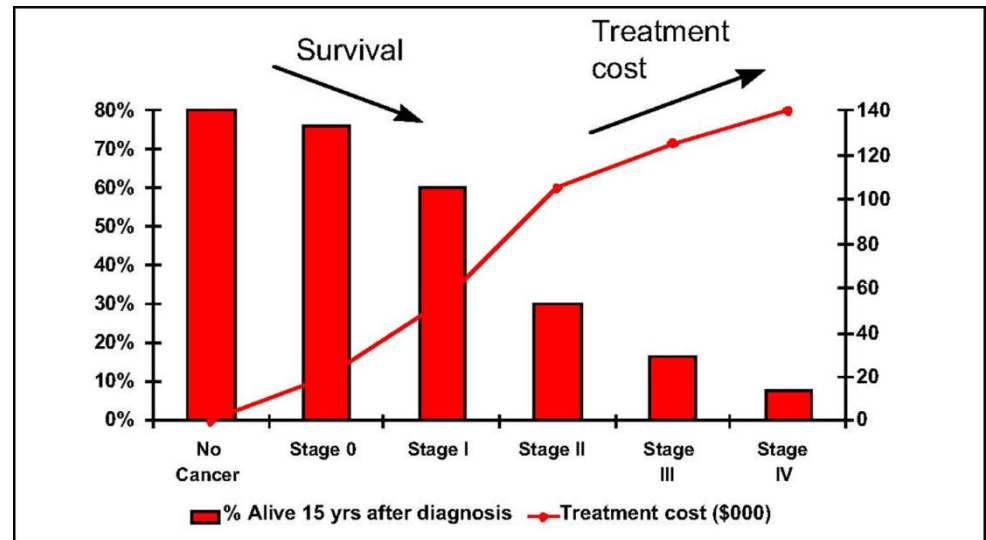
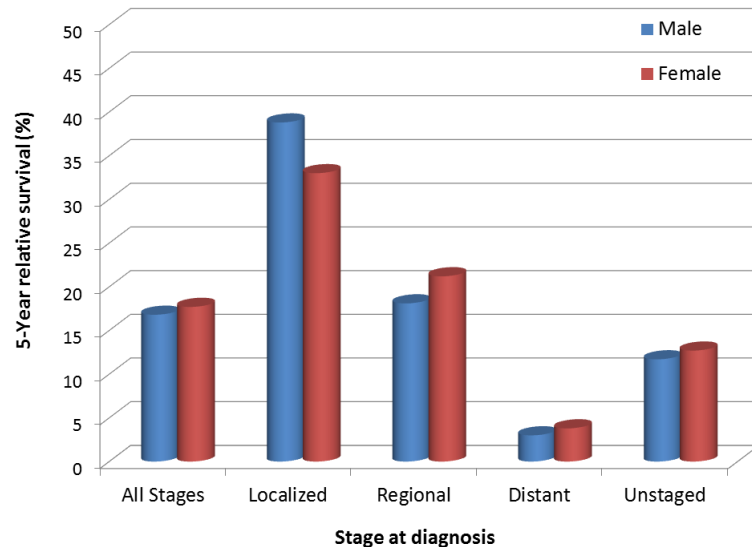
3% with cancer enroll in clinical trials



65% of patients would have been receptive to clinical trials



We need more effective treatments early on



Genetic testing to discover actionable mutations



- Personal genome sequencing

Germline or somatic mutations of an individual are found by comparing their sequence to a genome without mutations created in 'Human Genome Project'

Advances in biotechnology have decreased cost

- Purpose:

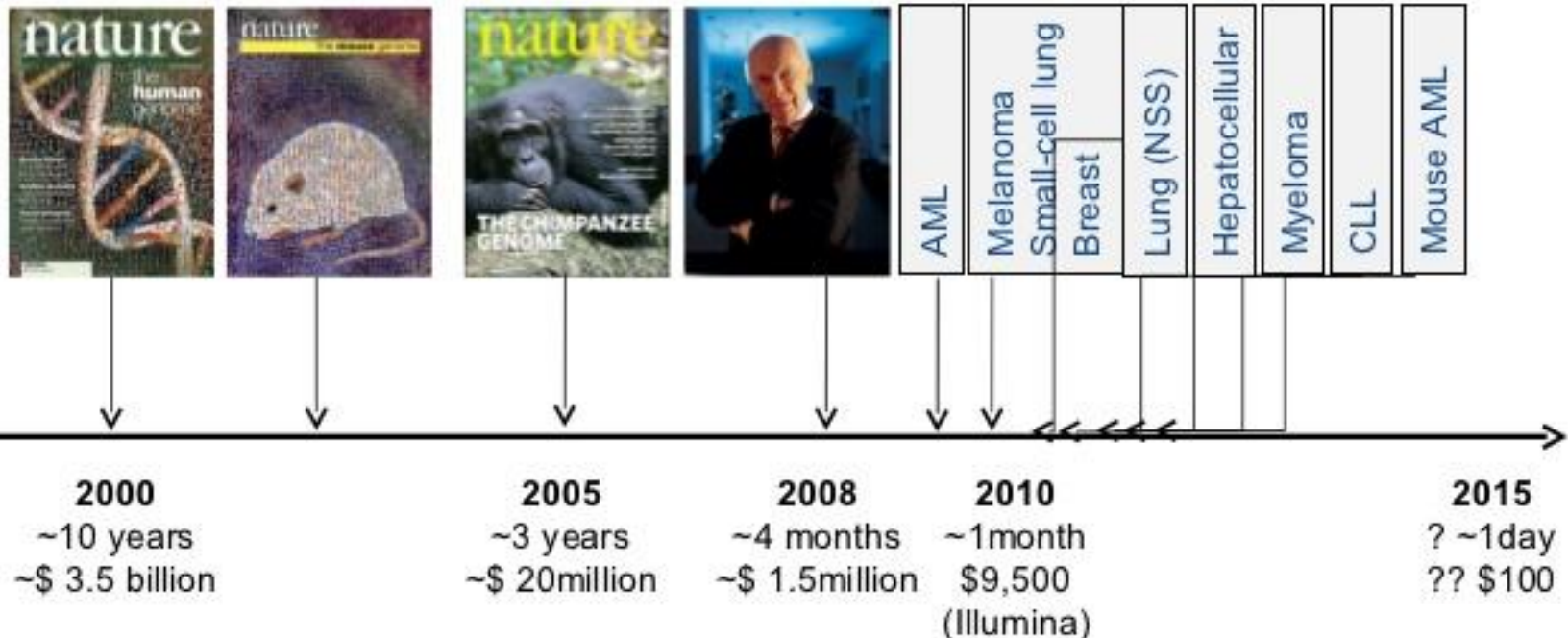
- 1. Examine determinants of hereditary cancer risk**
- 2. Discover therapeutic targets for personalized therapy**
- 3. Advance research in the search for the cure**
- 4. REPLACES INDIVIDUAL TESTS**

Genome Sequencing

Sanger (capillary) sequencing

Next generation sequencing

Cancer Genomics



Major testing categories

- **Hematological malignancies**
- **Solid tumors**
- **Inherited cancers**
- **Clinical trials: the first NOT the last stop in cancer treatment**

Focusing clinical trials on targeted subpopulations would slash their size, duration and cost

Facts and Figures

- 325,000 diagnosed every year in the UK
- 130,000 die from cancer
- 1.8 million living with cancer

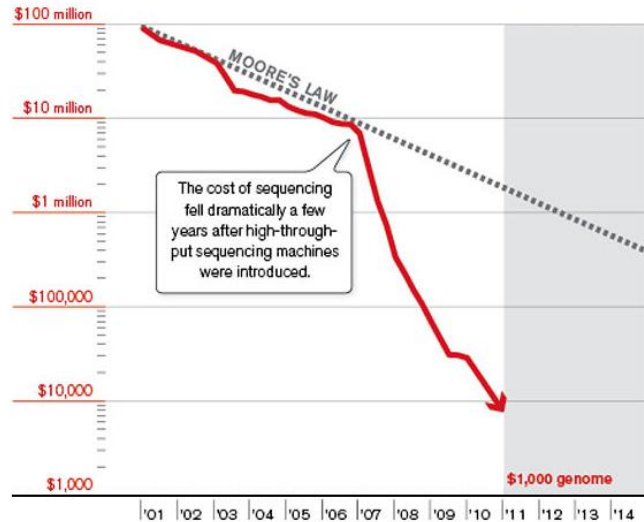
- £15.8bn total for all cancers
- £7.6bn economic costs
- £5.6bn for health
- £2.6bn for unpaid care

- Lung cancer £2.4bn per year
- Bowel cancer £1.6bn
- Breast cancer £1.5bn
- Prostate cancer £800m

Cost decreasing

Sequencing Costs Plummeting

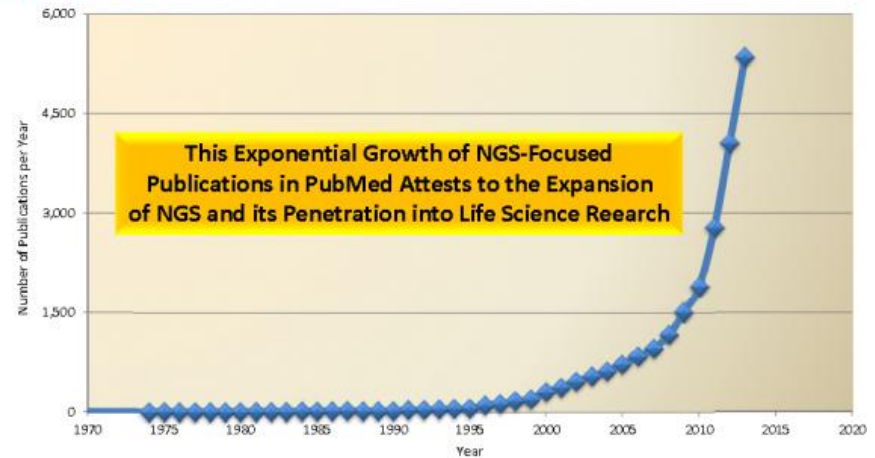
Cost per genome



Relevance increasing

GEN

Growth of Next-Generation Sequencing Focused Publications

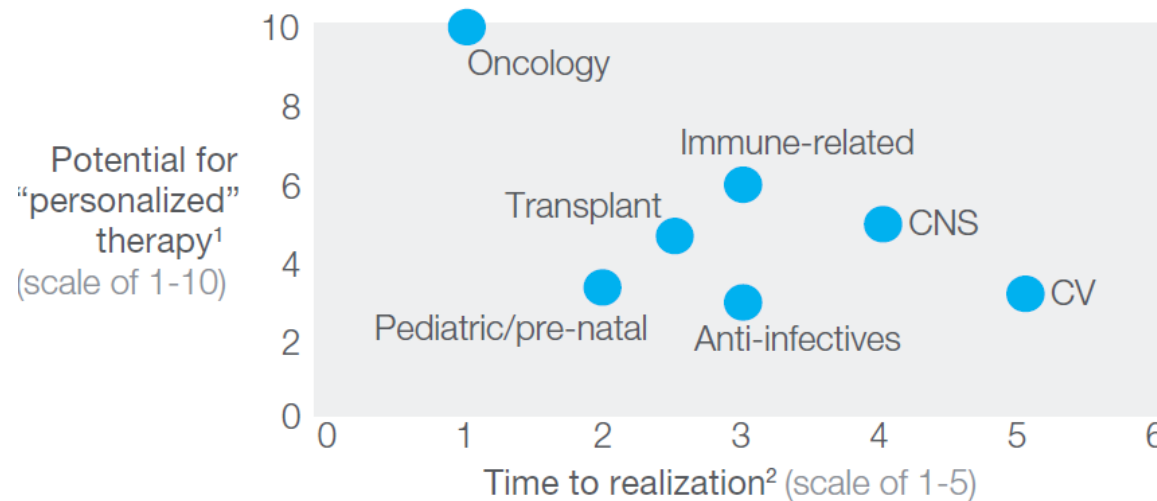


GENengnews.com

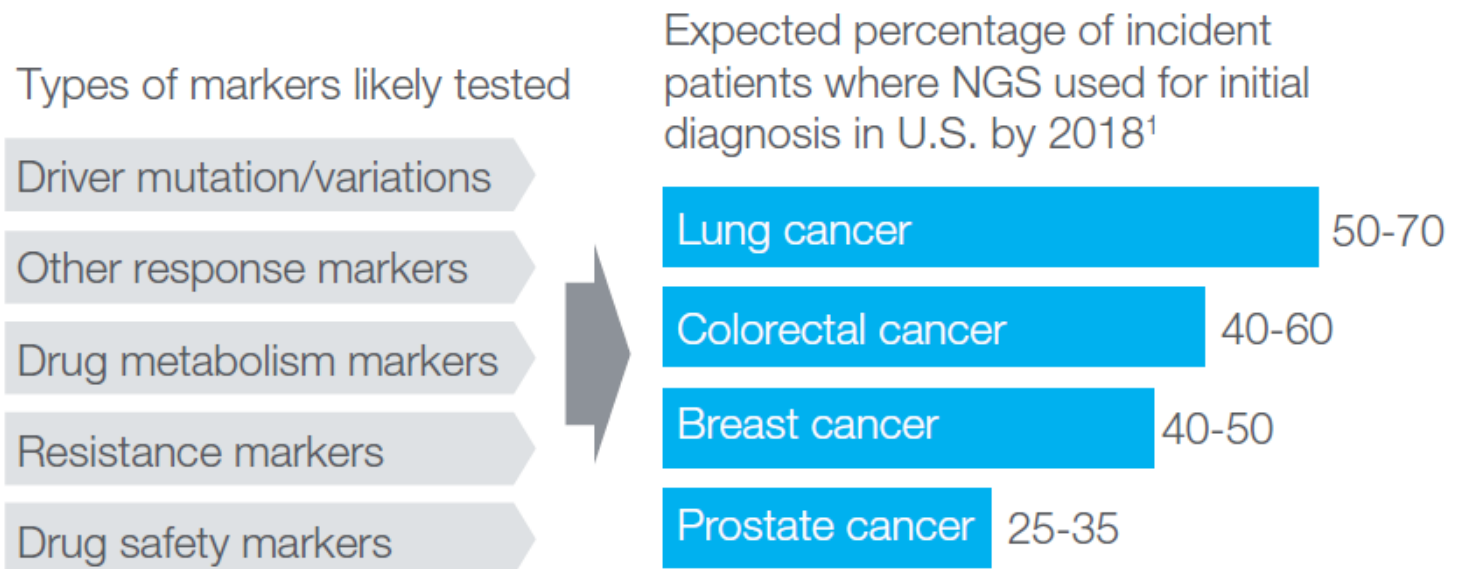
GEN Reports: Market & Tech Analysis, Produced by Enal Razvi, Ph.D., © 2013

In 2010, the average cost of treating someone diagnosed with cancer was approximately £30,000; by 2021 this may rise to a > £50,000.

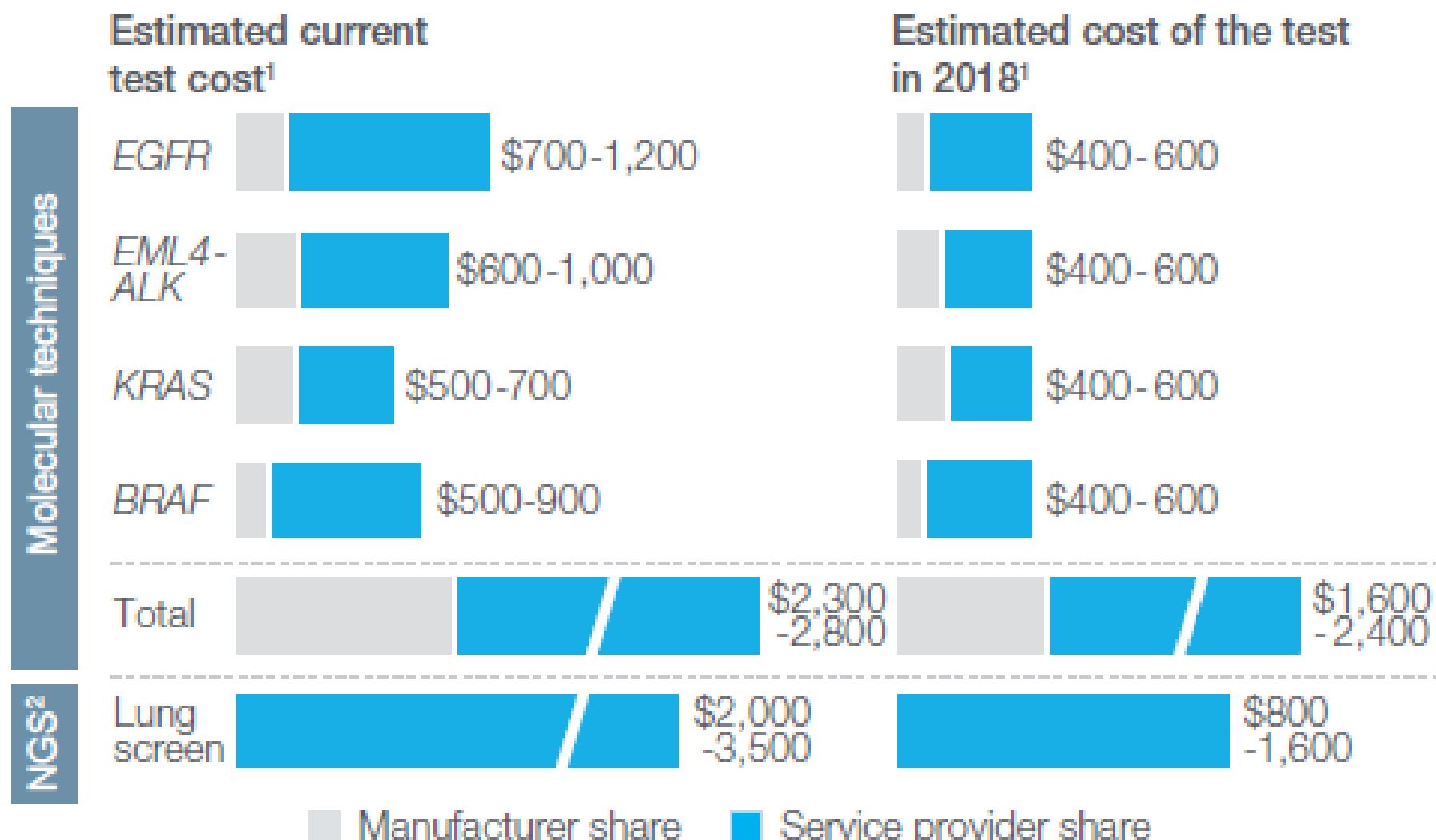
Potential for personalized medicine across disease areas



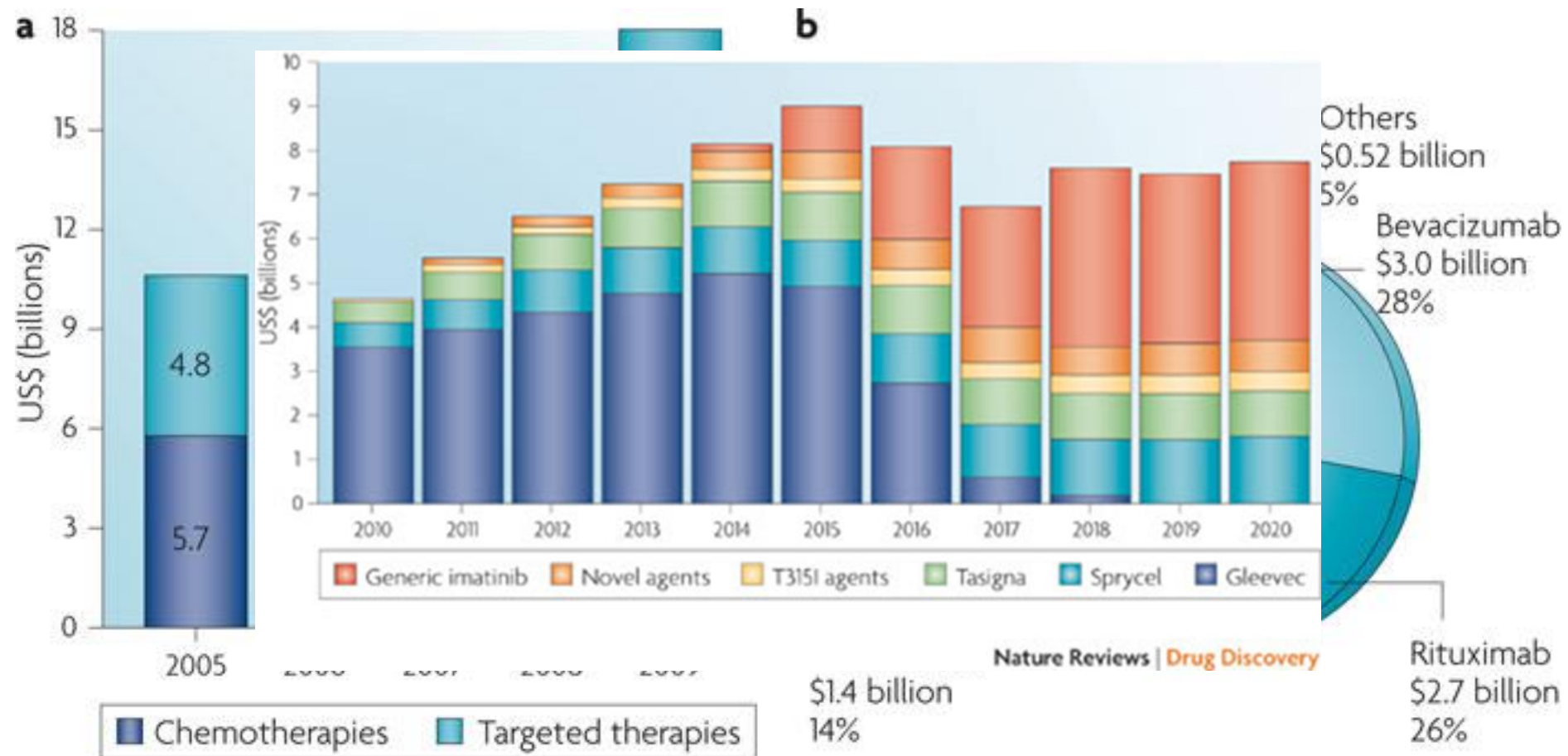
Relative penetration of next generation sequencing in Oncology in the future



Potential opportunity for panel testing—lung cancer example



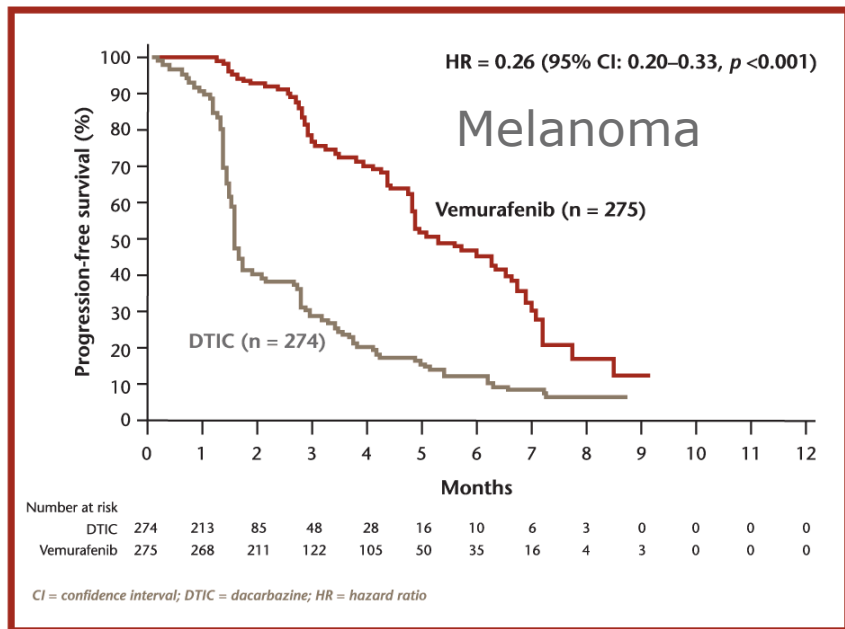
TARGETED TREATMENTS IN ONCOLOGY



Drug	Indication	Mechanism
<u>Imatinib mesylate (Gleevec®)</u>	• CML • gastrointestinal stromal tumor • dermatofibrosarcoma protuberans • myelodysplastic • myeloproliferative disorders • systemic mastocytosis	TKI bcr-abl
<u>Dasatinib (Sprycel®)</u>	• CML • acute lymphoblastic leukemia	TKI bcr-abl T315I
<u>Nilotinib (Tasigna®)</u>	• CML	TKI bcr-abl T315I
<u>Bosutinib (Bosulif®)</u>	• CML	TKI bcr-abl T315I
<u>Trastuzumab (Herceptin®)</u>	• breast cancer • gastroesophageal junction adenocarcinoma	Anti-HER2 mAb
<u>Pertuzumab (Perjeta™)</u>	• In combination with trastuzumab and docetaxel for metastatic breast cancer	Anti-HER2 mAb
<u>Lapatinib (Tykerb®)</u>	• advanced or metastatic breast cancer	TKI EGFR-HER2
<u>Gefitinib (Iressa®)</u>	• non-small cell lung cancer	TKI EGFR
<u>Erlotinib (Tarceva®)</u>	• metastatic non-small cell lung cancer and pancreatic cancer	TKI EGFR
<u>Cetuximab (Erbix®)</u>	• squamous cell carcinoma of the head and neck or colorectal cancer	mAb EGFR
<u>Panitumumab (Vectibix®)</u>	• metastatic colon cancer	mAb EGFR
<u>Temsirolimus (Torisel®)</u>	• advanced renal cell carcinoma	
<u>Everolimus (Afinitor®)</u>	• advanced kidney cancer • subependymal giant cell astrocytoma *TS • advanced breast cancer • pancreatic neuroendocrine tumors	mTOR

Drug	Indication	Mechanism
<u>Bortezomib (Velcade®)</u>	- multiple myeloma - mantle cell lymphoma	Proteasome inh.
<u>Carfilzomib (Kyprolis™)</u>	multiple myeloma	Proteasome inh.
<u>Pralatrexate (Folotyn®)</u>	peripheral T-cell lymphoma	Anti-folate
<u>Bevacizumab (Avastin®)</u>	- glioblastoma - non-small cell lung cancer - metastatic colorectal cancer - metastatic kidney cancer	Anti-VEGF mAb
<u>Ziv-aflibercept (Zaltrap®)</u>	metastatic colorectal cancer	Rec. fusion protein binding VEGF
<u>Sorafenib (Nexavar®)</u>	- advanced renal cell carcinoma - hepatocellular carcinoma	Multi TKI
<u>Sunitinib (Sutent®)</u>	- metastatic renal cell carcinoma - gastrointestinal stromal tumor - pancreatic neuroendocrine tumors	Multi TKI
<u>Pazopanib (Votrient®)</u>	- advanced renal cell carcinoma - soft tissue sarcoma	TKI VEGF/c-Kit/PDGFR
<u>Regorafenib (Stivarga®)</u>	metastatic colorectal cancer	TKI VEGF /RET/MET/TRKB/TIE2
<u>Cabozantinib (Cometriq™)</u>	metastatic medullary thyroid cancer	Anti-CD20 mAb
<u>Rituximab (Rituxan®)</u>	- B-cell non-Hodgkin lymphoma - chronic lymphocytic leukemia (CLL)	Anti-CD52 mAb
<u>Alemtuzumab (Campath®)</u>	B-cell CLL	Anti-CD20 mAb
<u>Ofatumumab (Arzerra®)</u>	CLL	Anti-CD20 mAb
<u>Ipilimumab (Yervoy™)</u>	unresectable or metastatic melanoma	Anti-CTLA4 mAb
<u>Tositumomab and 131I-tositumomab (Bexxar®)</u>	B-cell non-Hodgkin lymphoma	Anti-CD20-Iodine131
<u>Ibritumomab tiuxetan (Zevalin®)</u>	B-cell non-Hodgkin lymphoma	Anti-CD20-indium111/yttrium91
<u>Denileukin diftitox (Ontak®)</u>	CTCL	IL2-diphtheria toxin
<u>Brentuximab vedotin (Adcetris®)</u>	- anaplastic large cell lymphoma - Hodgkin lymphoma	Anti-CD30-monomethyl auristatin E
<u>T-DM1 (Kadcyla)</u>	Metastatic breast cancer	Trastuzumab-emtansine

Figure 3. BRIM-3 progression-free survival⁴⁵



A Progression-free Survival

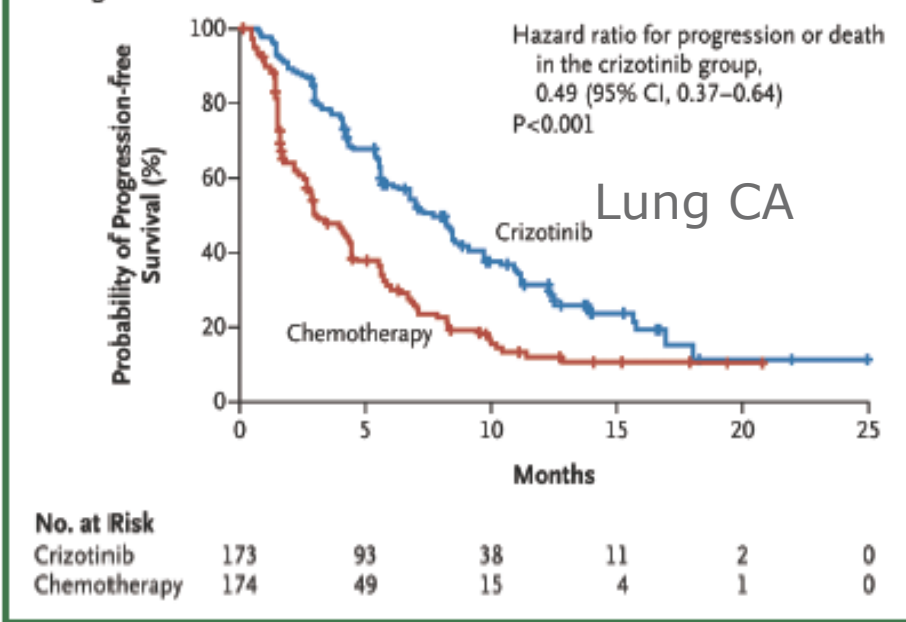
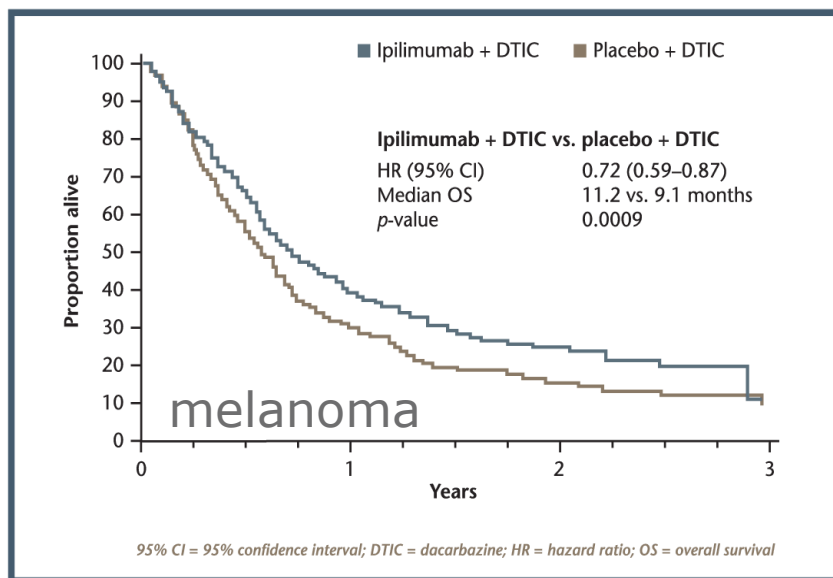
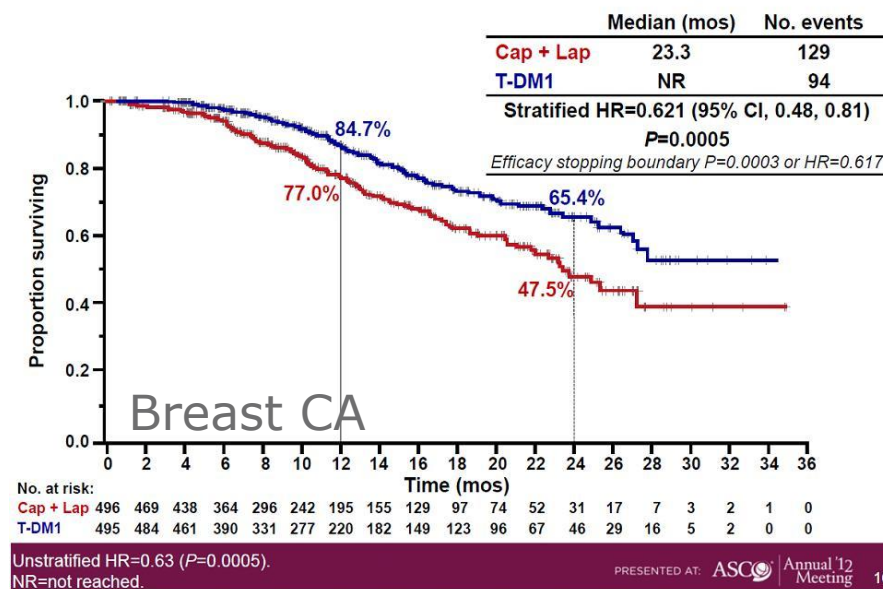


Figure 1. Overall survival



Overall Survival: Interim Analysis



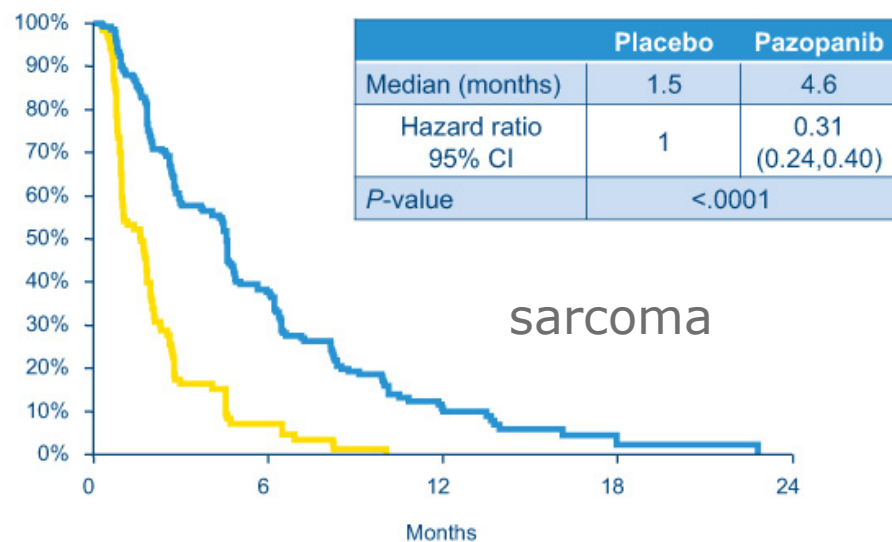
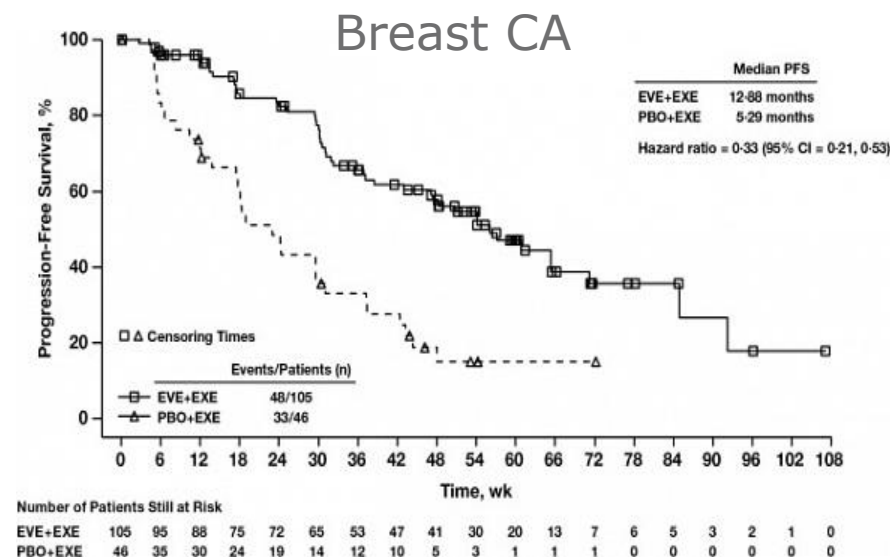
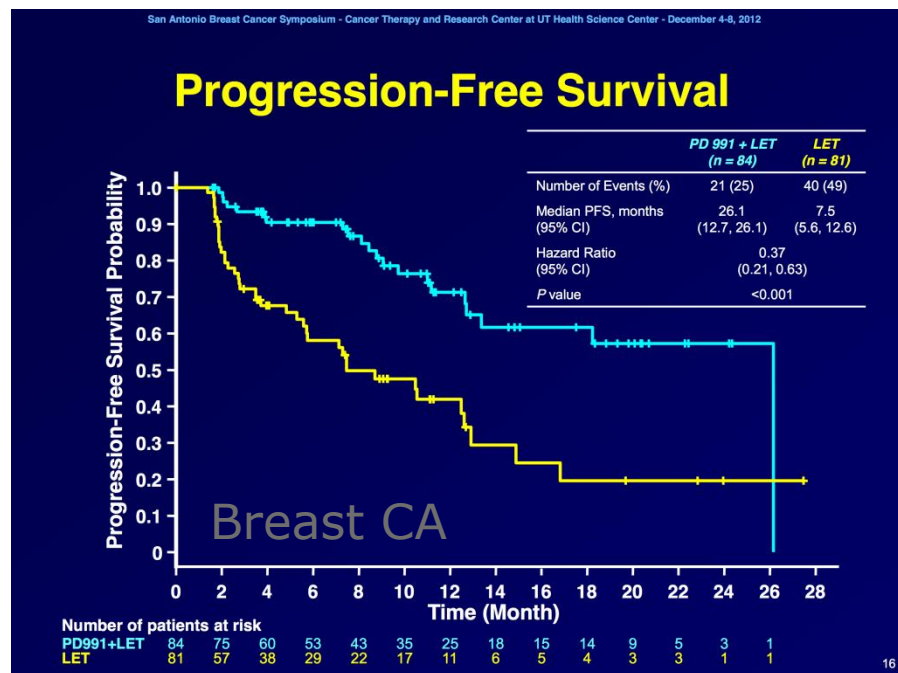
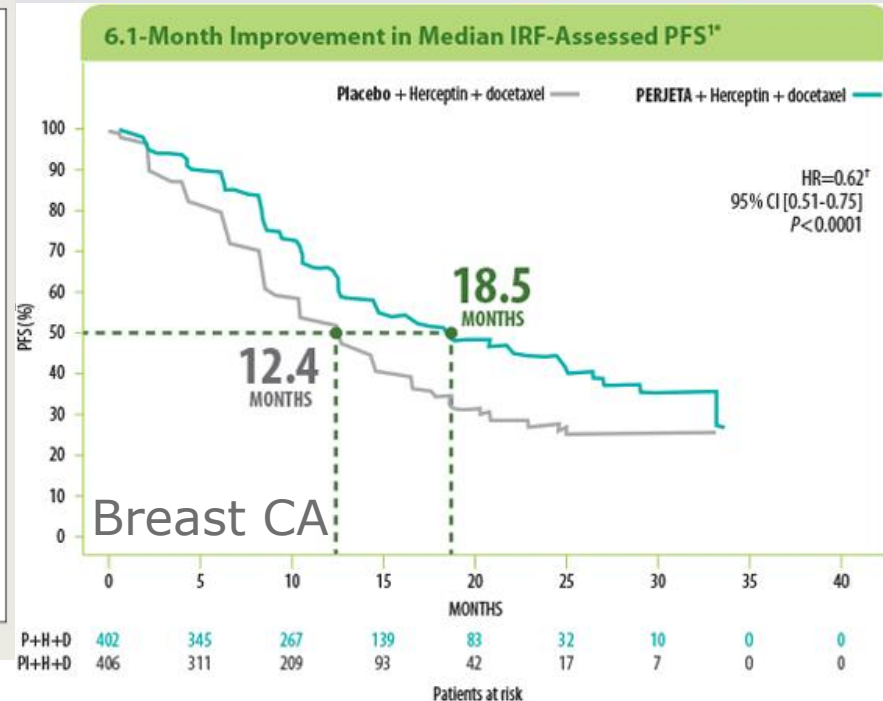


Fig. 1: Progression-free survival in the PALETTE trial. Courtesy of William D. Tap, MD. Adapted from Van Der Graaf WT, et al.³

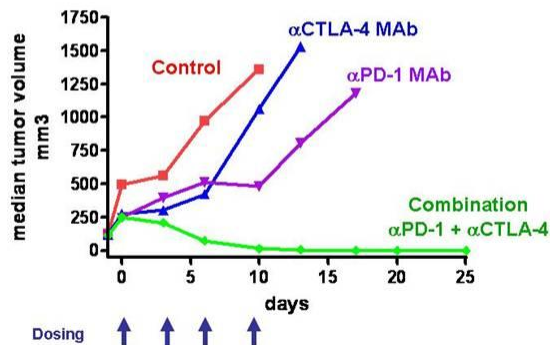


Immuno-therapies

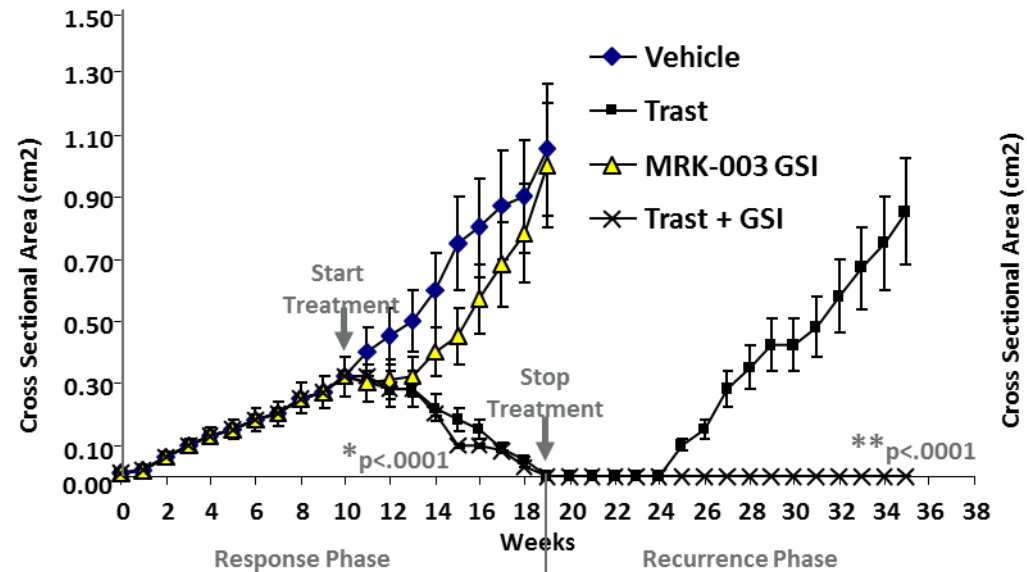
Cancer stem cell therapy

Synergistic Activity with Anti-PD-1 and Anti-CTLA-4 Antibodies

Combination of Sub-Efficacious Doses of anti-PD1 and anti-CTLA-4 Antibodies is Efficacious in Mouse Model



MEZAREX



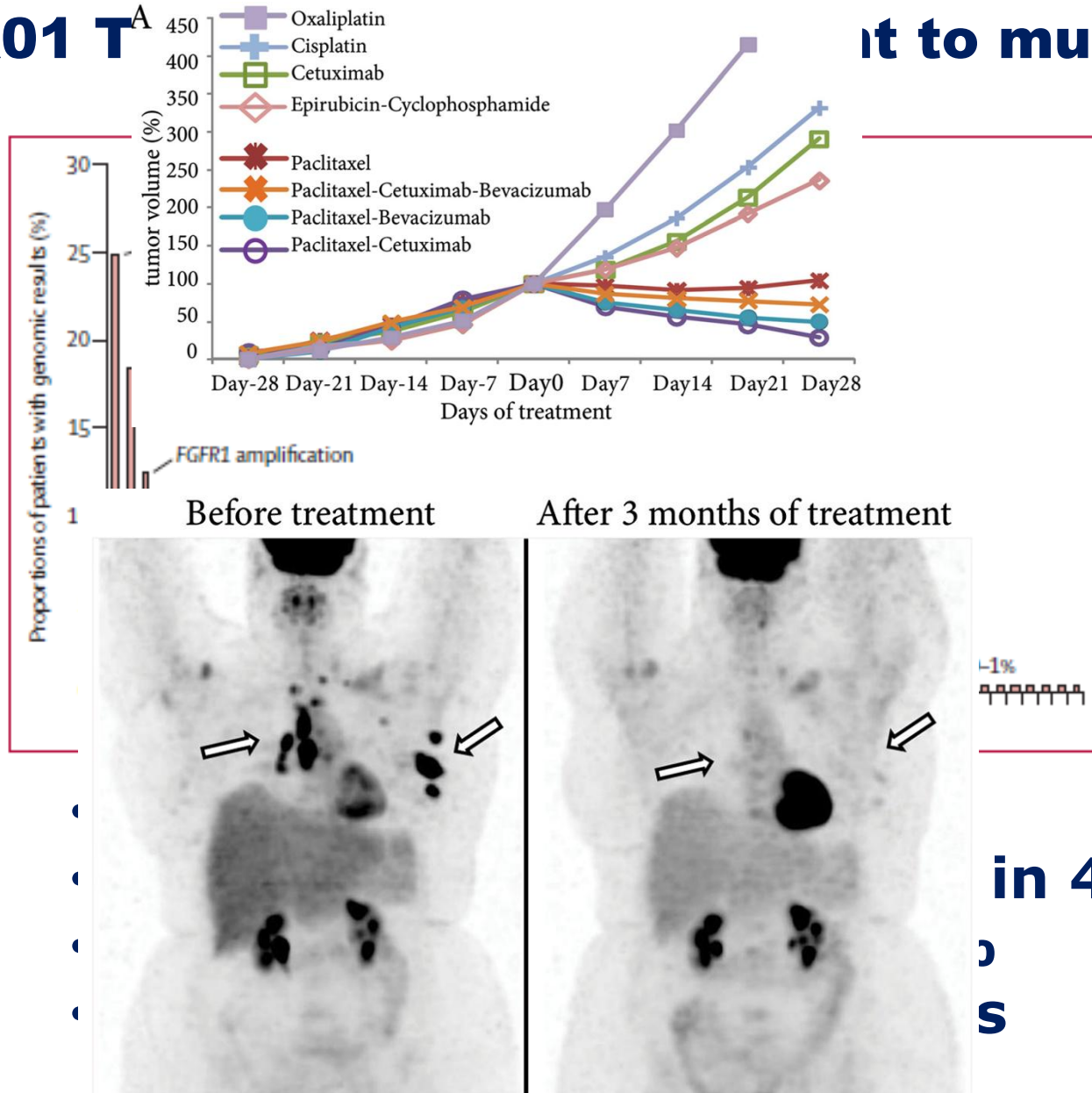
44 yr old TNBC patient

Molecular profiling revealed EGFR and KRAS wild type

Patient derived xenograft models based on genetics

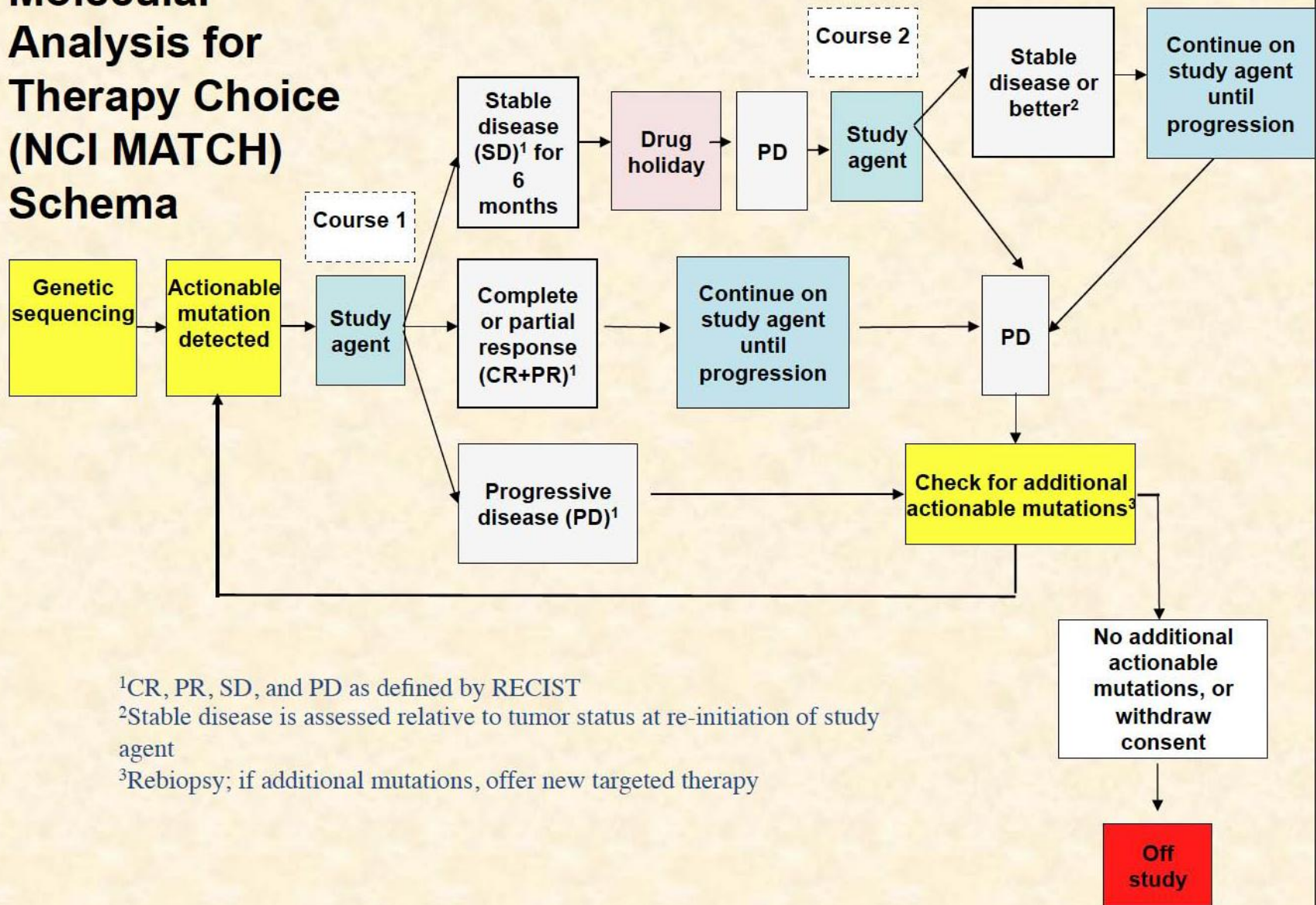
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it to mutations



The big initiative: NCI MATCH and UK MATCH Trials

Molecular Analysis for Therapy Choice (NCI MATCH) Schema



The Cancer Drug Fund

- Government Funding 200 million per year
- The CDF may be used for molecular diagnostic testing which is necessary to help optimally target the use of drugs to patients who are most likely to benefit
- Targeted treatments

Development and acceptance of personalized medicine

- Per patient savings
 - Likelihood that a test suggests an intervention for any particular patient
 - Minimizing costly adverse events or expensive procedures
 - BRCA: £20,000 saving when performed on adequate patients (family history, personal risk factors)
 - For some cases £20-30 K savings in treatment cost
- Example: Cetuximab in NRAS mutant tumours with efficacy of 7% vs 40% in wt NRAS

Incentive for adopting personalized medicine

- Power to transform health care from a population-based to an individual model.
- Design of therapies targeted to specific individuals most likely to benefit

Personalized medicine is not a promise of the future, it is fast emerging as the current state in diagnostics and therapeutics

ACKNOWLEDGEMENTS

Professor Justin Stebbing

From the laboratory to the clinic

As clichéd as it may sound, I chose to practice medicine to help people. In the clinic, ones job is well defined; but the world of research is infinite. There is no end to it, including the work one needs to put in to obtain a result.

[Read on](#)



**We have to work
together
to find treatments
that will cure cancer**

